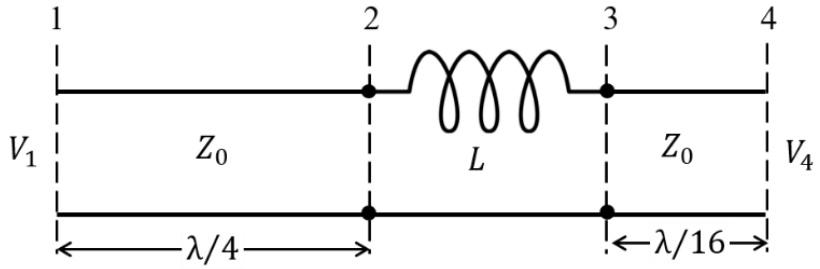
Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 1 (06.11.2020)

Problem 1.1: Scattering parameters

The circuit, as shown below, consists of two transmission lines with different lengths connected to an inductor. Let $\omega L = 100\Omega$ and the characteristic impedance of the transmission lines $Z_0 = 50\Omega$.



(a) Calculate the reflection coefficients at port 2 and port 3 looking towards the inductor L.
(b) With these, give the corresponding S-parameters of the circuit between ports 2 and 3.
(c) Determine the scattering matrix of the whole circuit between ports 1 and 4.
(d) By means of this scattering matrix, show that the circuit is lossless.
(e) What is power transmission ratio (P4 / P1) from port 1 to port 4?

Let a load $Z_L = 292\Omega$ be connected to port 4.

(f) What is the load reflection coefficient at port 4?
(g) Calculate the input reflection coefficient at port 1.
(h) Determine the voltage transformation ratio V4 / V1.

Problem 1.2: Scattering matrix

A four-port network has the scattering matrix shown as follows:

$$\begin{bmatrix} 0.178\angle 90° & 0.6\angle 45° & 0.4\angle 45° & 0 \\ 0.6\angle 45° & 0 & 0 & 0.3\angle -45° \\ 0.4\angle 45° & 0 & 0 & 0.5\angle -45° \\ 0 & 0.3\angle -45° & 0.5\angle -45° & 0 \end{bmatrix}$$

(a) Is this network lossless?
(b) Is this network reciprocal?
(c) What is the return loss at port 1 when all other ports are terminated with matched loads?
(d) What is the insertion loss between ports 2 and 4 when all other ports are terminated with matched loads?
(e) What is the phase delay between ports 2 and 4 when all other ports are terminated with matched loads?
(f) What is the reflection coefficient seen at port 1 if a short circuit is placed at the terminal plane of port 3 and all other ports are terminated with matched loads?